

STUDY UNIT 7

MORE ABOUT FUNCTIONS

Complete taught notes: range, injectivity, surjectivity, composition, bijections, inverse functions and identity functions.

✓ Correct symbols	✓ Proof templates
✓ Example per topic	✓ Mapping diagrams
✓ Exam traps	✓ Practice + solutions

THE MASTER IDEA

A function is **bijective** when it has no collisions and no missing codomain values. Exactly then, every output has one unique input, so all arrows can be reversed to form an inverse function.

How to Use These Notes

This unit is easiest when you separate three questions:

Question	Concept	Fast memory
What values actually come out?	Range	Actual outputs
Does every allowed output get reached?	Surjective	No empty spots
Can different inputs collide?	Injective	No collisions

EXAM HABIT

Always write the **domain** and **codomain** beside the formula before deciding whether a function is injective, surjective, bijective or invertible. The formula alone is not enough.

Complete contents

1. Function foundations: relations, functions, domain, codomain, images and preimages
2. Range and determining range
3. Surjective (onto) functions and proofs
4. Injective (one-to-one) functions and proofs
5. Comparing the four function types
6. Composition of functions
7. Theorems about composition
8. Bijective functions and finite-set shortcuts
9. Inverse relations and inverse functions
10. Finding and verifying inverse functions
11. Domain restriction and inverse graphs
12. Identity functions and inverse laws
13. Integrated exam examples
14. Common traps
15. Practice test
16. Fully worked solutions
17. Final memory sheet and mastery checklist

1. Function Foundations

Before the new Unit 7 properties make sense, the basic language must be exact.

1.1 Relations and functions

A **relation** from A to B is any subset of $A \times B$. A **function** $f: A \rightarrow B$ is a relation in which every element of A occurs as a first coordinate exactly once.

- Every domain element must have an output.
- One input may not have two different outputs.
- Different inputs may share an output; that is allowed, although the function will not be injective.

WORKED EXAMPLE 1.1

Let $A = \{1, 2, 3\}$ and $R = \{(1, a), (2, b), (3, a)\}$. Every member of A appears once as a first coordinate, so R is a function. The repeated output a is allowed.

$S = \{(1, a), (1, b), (2, a), (3, b)\}$ is not a function because input 1 has two outputs.

1.2 Domain, codomain, image and preimage

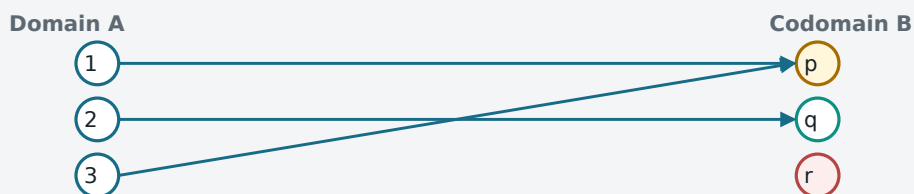
For $f: A \rightarrow B$, A is the **domain** and B is the **codomain**. If $f(a) = b$, then b is the **image** of a , while a is a **preimage** of b . An output can have zero, one or several preimages.

$$f : A \rightarrow B \quad a \mapsto f(a)$$

WORKED EXAMPLE 1.2

For $f = \{(1, p), (2, q), (3, p)\}$ from $\{1, 2, 3\}$ to $\{p, q, r\}$: the image of 2 is q ; the preimages of p are 1 and 3; r has no preimage.

Function with a collision and an unused codomain value



EXTRA EXAMPLE

Let $h: \{a, b, c, d\} \rightarrow \{2, 4, 6\}$ be $h(a) = 2$, $h(b) = 4$, $h(c) = 4$, $h(d) = 6$. The image of c is 4, the preimages of 4 are b and c , and every domain element has exactly one output.

2. Range of a Function

The range records the values that are actually produced, not merely the values permitted by the codomain.

2.1 Definition

$$\text{ran}(f) = \{ f(x) \mid x \in A \} = f[A] \text{ and always } \text{ran}(f) \subseteq B$$

The range may equal the codomain, but it does not have to. Equality happens exactly when f is surjective.

WORKED EXAMPLE 2.1 - Finite function

If $f: \{1,2,3,4\} \rightarrow \{a,b,c,d,e\}$ is given by $1 \mapsto a$, $2 \mapsto c$, $3 \mapsto c$, $4 \mapsto e$, then $\text{ran}(f) = \{a,c,e\}$. The codomain is $\{a,b,c,d,e\}$; b and d are allowed but never produced.

2.2 Determining a range from ordered pairs or a table

Collect the distinct second coordinates. Repeated outputs are written once because a range is a set.

WORKED EXAMPLE 2.2

For $g = \{(-2,4), (-1,1), (0,0), (1,1), (2,4)\}$, $\text{ran}(g) = \{0,1,4\}$. Do not write repeated 1 and 4 twice.

2.3 Determining an algebraic range

- Write $y = f(x)$.
- Solve for x in terms of y when useful.
- Apply the domain restriction.
- State exactly which y -values are possible.

WORKED EXAMPLE 2.3 - Integers

Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = 4x + 5$. From $y = 4x + 5$ we get $x = (y - 5)/4$. This is an integer only when $y \equiv 1 \pmod{4}$. Therefore $\text{ran}(f) = \{4k + 1 \mid k \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\}$.

EXTRA EXAMPLE - Real square

For $q: \mathbb{R} \rightarrow \mathbb{R}$, $q(x) = x^2$, all outputs are non-negative and every $y \geq 0$ occurs by choosing $x = \sqrt{y}$. Thus $\text{ran}(q) = [0, \infty)$.

2.4 Range rules worth knowing

Function and domain	Range	Reason
x^2 on $\mathbb{R}$	$[0, \infty)$	Squares are non-negative
x^2 on $[-3, 2]$	$[0, 9]$	0 is minimum; $(-3)^2 = 9$ is maximum
$1/x$ on $\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	A reciprocal is never 0
$\sqrt{x}$ on $[0, \infty)$	$[0, \infty)$	Principal square root is non-negative
$mx + c$ on $\mathbb{R}$, $m \neq 0$	$\mathbb{R}$	Solve $x = (y - c)/m$ for every real y

EXTRA EXAMPLE - Restricted interval

For $p: [-2, 3] \rightarrow \mathbb{R}$, $p(x) = x^2$, the smallest output is 0 and the largest is 9, so $\text{ran}(p) = [0, 9]$. Notice that checking only endpoint values would miss the minimum at $x = 0$.

3. Surjective (Onto) Functions

Surjectivity compares the range with the declared codomain.

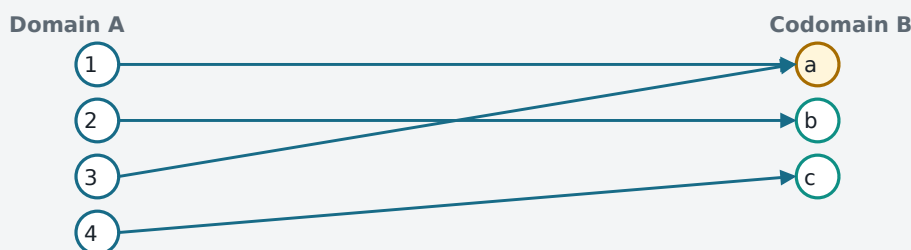
3.1 Definition and mapping test

$f:A \rightarrow B$ is surjective $\Leftrightarrow \text{ran}(f)=B \Leftrightarrow \forall y \in B, \exists x \in A$ such that $f(x)=y$

MEMORY

Surjective means **no empty spots** in the codomain. Several inputs may still hit the same output.

Surjective but not injective



WORKED EXAMPLE 3.1

The diagram reaches a, b and c, so the range equals the codomain and the function is surjective. It is not injective because 1 and 3 collide at a.

3.2 Standard proof of surjectivity

- Take an arbitrary y in the codomain.
- Solve $y=f(x)$ for x .
- Show this x belongs to the domain.
- Conclude that every y has a preimage.

WORKED EXAMPLE 3.2 - Proof

Prove $f:\mathbb{Z} \rightarrow \mathbb{Z}$, $f(x)=x-6$ is onto.

Let $y \in \mathbb{Z}$. Choose $x=y+6$. Since $y+6 \in \mathbb{Z}$ and $f(x)=(y+6)-6=y$, every integer y is reached. Therefore f is surjective.

3.3 Disproving surjectivity

One missing codomain element is enough. Give a specific $y \in B$ and show that $f(x)=y$ has no solution in A .

WORKED EXAMPLE 3.3 - Counterexample

For $f:\mathbb{Z} \rightarrow \mathbb{Z}$, $f(x)=2x+1$, choose $y=0$. Solving $2x+1=0$ gives $x=-1/2$, which is not an integer. Hence $0 \in \mathbb{Z}$ is not reached, so f is not surjective.

EXTRA EXAMPLE

For $s:\mathbb{R} \rightarrow \mathbb{R}$, $s(x)=x^2+4$, $\text{ran}(s)=[4, \infty)$. Since 0 belongs to the codomain but not the range, s is not surjective.

4. Injective (One-to-One) Functions

Injectivity controls collisions: different inputs must give different outputs.

4.1 Definition and equivalent forms

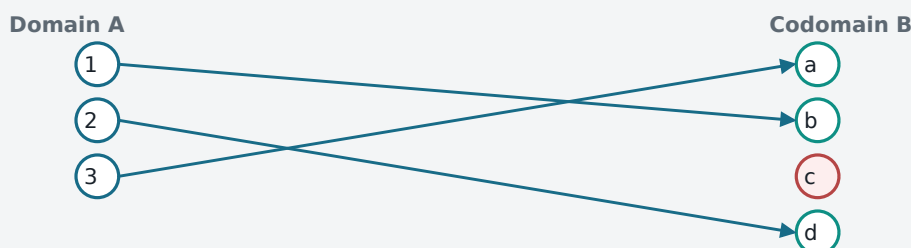
f is injective $\Leftrightarrow f(u)=f(v) \Rightarrow u=v$

Equivalent contrapositive: $u \neq v \Rightarrow f(u) \neq f(v)$

MEMORY

Injective means **no collisions**. Unused codomain values are allowed.

Injective but not surjective



WORKED EXAMPLE 4.1

All three outputs are different, so the function is injective. The unused value c does not affect injectivity.

4.2 Standard algebraic proof

- Let u, v be arbitrary elements of the domain.
- Assume $f(u)=f(v)$.
- Use algebra to force $u=v$.
- State the conclusion: f is injective.

WORKED EXAMPLE 4.2 - Proof

Prove $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x)=5x-7$ is one-to-one.

Assume $f(u)=f(v)$. Then $5u-7=5v-7$, so $5u=5v$ and $u=v$. Therefore f is injective.

4.3 Disproving injectivity

Find one collision: two distinct domain values with the same image.

WORKED EXAMPLE 4.3 - Counterexample

For $q: \mathbb{Z} \rightarrow \mathbb{Z}$, $q(x)=x^2$, $q(2)=4$ and $q(-2)=4$ although $2 \neq -2$. Therefore q is not injective.

4.4 Horizontal-line idea

For a real-valued graph, a function is injective when every horizontal line meets its graph at most once. This is a visual test, not a replacement for a formal proof when proof is requested.

EXTRA EXAMPLE

The graph $y=x^3$ is passed by every horizontal line once, so $x^3: \mathbb{R} \rightarrow \mathbb{R}$ is injective. The graph $y=x^2$ is hit twice by many horizontal lines, so $x^2: \mathbb{R} \rightarrow \mathbb{R}$ is not injective.

5. The Four Function Types

Injectivity and surjectivity are independent properties. A function can have neither, either one, or both.

Type	Collision?	Missing codomain value?	Example
Neither	Yes	Yes	$x^2: \mathbb{Z} \rightarrow \mathbb{Z}$
Injective only	No	Yes	$4x+1: \mathbb{Z} \rightarrow \mathbb{Z}$
Surjective only	Yes	No	$\{1,2,3\} \rightarrow \{a,b\}: 1 \mapsto a, 2 \mapsto b, 3 \mapsto a$
Bijjective	No	No	$x+1: \mathbb{Z} \rightarrow \mathbb{Z}$

WORKED EXAMPLE 5.1

Classify $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x)=3x$.

Injective: $3u=3v$ implies $u=v$. Not surjective: 1 is not a multiple of 3, so no integer maps to 1. Hence f is injective only.

EXTRA EXAMPLE

Let $A=\{1,2,3,4\}$, $B=\{p,q,r\}$, with $1 \mapsto p$, $2 \mapsto q$, $3 \mapsto r$, $4 \mapsto p$. Every codomain element is reached, but 1 and 4 collide. The function is surjective only.

5.1 Pigeonhole shortcuts for finite sets

$|A| > |B| \Rightarrow$ injective $A \rightarrow B$ impossible

$|A| < |B| \Rightarrow$ surjective $A \rightarrow B$ impossible

If $|A|=|B| < \infty$, then injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective

WORKED EXAMPLE 5.2

If $|A|=7$ and $|B|=5$, every function $A \rightarrow B$ must have a collision, so no such function can be injective. Some functions can still be surjective.

6. Composition of Functions

Composition connects two compatible functions to create a new function.

6.1 Definition and type compatibility

If $f:A \rightarrow B$ and $g:B \rightarrow C$, then $g \circ f:A \rightarrow C$ and $(g \circ f)(x) = g(f(x))$.

ORDER RULE

Read from right to left: in $g \circ f$, apply **f first**, then apply **g**.

More generally, $g \circ f$ is defined when every output of f is a valid input for g ; formally $\text{ran}(f) \subseteq \text{dom}(g)$.

WORKED EXAMPLE 6.1 - Formula

Let $f(x) = 2x + 1$ and $g(x) = x^2$. Then $(g \circ f)(x) = g(2x + 1) = (2x + 1)^2$, while $(f \circ g)(x) = f(x^2) = 2x^2 + 1$. They differ.

6.2 Composition from ordered pairs

WORKED EXAMPLE 6.2

Let $f = \{(1, a), (2, c), (3, b)\}$ and $g = \{(a, 10), (b, 20), (c, 30)\}$. Follow each chain:
 $1 \rightarrow a \rightarrow 10$, $2 \rightarrow c \rightarrow 30$, $3 \rightarrow b \rightarrow 20$.
 Therefore $g \circ f = \{(1, 10), (2, 30), (3, 20)\}$.

6.3 Composition is generally not commutative

Usually $g \circ f \neq f \circ g$

EXTRA EXAMPLE

Let $f(x) = x + 4$ and $g(x) = 2x$. Then $(g \circ f)(x) = 2x + 8$, whereas $(f \circ g)(x) = 2x + 4$. Order matters.

6.4 Why composition is a function

Each $a \in A$ has exactly one image $f(a) \in B$. That element has exactly one image $g(f(a)) \in C$. Therefore each a has exactly one final output under $g \circ f$.

EXTRA EXAMPLE

If f sends each student number to exactly one student and g sends each student to exactly one programme, then $g \circ f$ sends each student number to exactly one programme.

6.5 Domains of algebraic compositions

A composition is permitted only where x lies in $\text{dom}(f)$ and $f(x)$ lies in $\text{dom}(g)$.

WORKED EXAMPLE 6.5

Let $f(x) = x - 2$ and $g(x) = \sqrt{x}$. Then $(g \circ f)(x) = \sqrt{x - 2}$, so $x - 2 \geq 0$ and the domain is $[2, \infty)$. But $(f \circ g)(x) = \sqrt{x} - 2$ has domain $[0, \infty)$.

7. Theorems About Composition

These results explain what properties survive when functions are chained.

7.1 Composition of injective functions

f and g injective $\Rightarrow g \circ f$ injective

Proof: assume $(g \circ f)(u) = (g \circ f)(v)$. Then $g(f(u)) = g(f(v))$. Since g is injective, $f(u) = f(v)$. Since f is injective, $u = v$.

EXAMPLE 7.1

$f(x) = 3x$ and $g(x) = x - 5$ are injective $\mathbb{R} \rightarrow \mathbb{R}$. Their composition $(g \circ f)(x) = 3x - 5$ is therefore injective.

7.2 Composition of surjective functions

f and g surjective $\Rightarrow g \circ f$ surjective

Proof: take $c \in C$. Since g is onto, $c = g(b)$ for some $b \in B$. Since f is onto, $b = f(a)$ for some $a \in A$. Thus $c = g(f(a))$, so c is reached by $g \circ f$.

EXAMPLE 7.2

$f(x) = x + 1$ and $g(x) = 2x$ are surjective $\mathbb{R} \rightarrow \mathbb{R}$. Hence $(g \circ f)(x) = 2x + 2$ is surjective.

7.3 Composition of bijections

f and g bijective $\Rightarrow g \circ f$ bijective

EXAMPLE 7.3

$f(x) = x - 4$ and $g(x) = -2x + 3$ are bijections $\mathbb{R} \rightarrow \mathbb{R}$, so $(g \circ f)(x) = -2x + 11$ is also bijective.

7.4 Cancellation consequences

- If $g \circ f$ is injective, then f must be injective.
- If $g \circ f$ is surjective, then g must be surjective.
- The reverse-looking statements are not guaranteed: $g \circ f$ injective need not make g injective on all of its domain; $g \circ f$ surjective need not make f surjective onto all of its codomain.

EXAMPLE 7.4

If $g \circ f$ is injective but $f(u) = f(v)$, applying g gives $g(f(u)) = g(f(v))$, contradicting injectivity unless $u = v$. Therefore f must be injective.

7.5 Associativity

$h \circ (g \circ f) = (h \circ g) \circ f$

Both sides apply f first, then g , then h . Composition is associative even though it is usually not commutative.

EXAMPLE 7.5

Let $f(x) = x + 1$, $g(x) = 2x$, $h(x) = x^2$. Both $h \circ (g \circ f)$ and $(h \circ g) \circ f$ equal $(2x + 2)^2$.

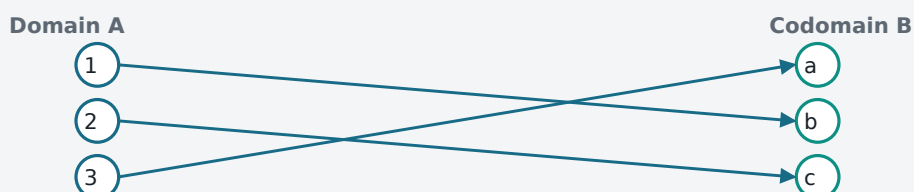
8. Bijective Functions

A bijection is simultaneously one-to-one and onto.

bijection = injective + surjective

A bijection pairs every domain element with a unique codomain element and leaves nothing unmatched.

A bijection: perfect matching



WORKED EXAMPLE 8.1 - Full proof

Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x + 7$.

Injective: if $u + 7 = v + 7$, then $u = v$.

Surjective: for any $y \in \mathbb{Z}$, choose $x = y - 7 \in \mathbb{Z}$; then $f(x) = y$.
Therefore f is bijective.

8.1 Counting finite bijections

If finite sets A and B each contain n elements, then the number of bijections $A \rightarrow B$ is $n!$ because the first input has n choices, the next $n-1$, and so on.

Number of bijections between two n -element sets = $n!$

EXAMPLE 8.2

If $|A| = |B| = 4$, there are $4! = 4 \times 3 \times 2 \times 1 = 24$ bijections $A \rightarrow B$.

EXTRA EXAMPLE - Impossible case

No bijection exists from a 3-element set to a 5-element set because a finite bijection requires equal cardinalities.

9. Inverse Relations and Inverse Functions

An inverse relation reverses every ordered pair. It becomes an inverse function only under the right conditions.

9.1 Inverse relation

If $R \subseteq A \times B$, then $R^{-1} = \{ (b, a) \mid (a, b) \in R \}$.

EXAMPLE 9.1

If $R = \{(1, a), (2, b), (3, c)\}$, then $R^{-1} = \{(a, 1), (b, 2), (c, 3)\}$.

9.2 Why injectivity is necessary

If two inputs of f share an output, reversing the pairs gives one inverse input two outputs, violating the definition of a function.

EXAMPLE 9.2

$f = \{(1, a), (2, a), (3, b)\}$ gives $f^{-1} = \{(a, 1), (a, 2), (b, 3)\}$. Since a has two outputs, f^{-1} is not a function.

9.3 Why surjectivity is necessary

If some $b \in B$ is not reached, then after reversal b has no output. Thus the inverse cannot be a function with domain equal to all of B .

EXAMPLE 9.3

For $f: \{1, 2\} \rightarrow \{a, b, c\}$, $1 \mapsto a$ and $2 \mapsto b$, the value c has no preimage. After reversal, c has no output, so there is no inverse function $\{a, b, c\} \rightarrow \{1, 2\}$.

9.4 Fundamental theorem

f is invertible $\Leftrightarrow f$ is bijective

If f is bijective, each $b \in B$ has exactly one preimage $a \in A$, so reversing all arrows produces a function $f^{-1}: B \rightarrow A$. Conversely, if f^{-1} is a function on all of B , every b must have exactly one preimage, making f both onto and one-to-one.

EXAMPLE 9.4

$f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = 2x + 1$ is injective but not surjective because it reaches only odd integers. Therefore it is not bijective and has no inverse function $\mathbb{Z} \rightarrow \mathbb{Z}$.

NOTATION TRAP

$f^{-1}(x)$ means the inverse function. It does **not** mean $1/f(x)$. For example, if $f(x) = 2x$, then $f^{-1}(x) = x/2$, while $1/f(x) = 1/(2x)$.

10. Finding and Verifying Inverse Functions

Only declare an inverse function after confirming that the original function is bijective on its stated domain and codomain.

10.1 Algebraic method

- Write $y=f(x)$.
- Interchange x and y .
- Solve for y .
- Write $y=f^{-1}(x)$.
- Swap domain and codomain, and verify if required.

WORKED EXAMPLE 10.1

Let $f:\mathbb{R}\rightarrow\mathbb{R}$, $f(x)=3x-5$. It is a non-constant linear function, hence bijective. Write $y=3x-5$; swap: $x=3y-5$; solve: $y=(x+5)/3$. Therefore $f^{-1}(x)=(x+5)/3$.

10.2 Verification by composition

$$f^{-1}\circ f = i_A \text{ and } f\circ f^{-1} = i_B$$

WORKED EXAMPLE 10.2

For $f(x)=3x-5$ and $f^{-1}(x)=(x+5)/3$:
 $(f^{-1}\circ f)(x)=((3x-5)+5)/3=x$;
 $(f\circ f^{-1})(x)=3((x+5)/3)-5=x$.
 Both identity compositions hold.

10.3 Self-inverse functions

A function is self-inverse when $f^{-1}=f$, equivalently $f(f(x))=x$.

EXAMPLE 10.3

Let $f:\mathbb{R}\rightarrow\mathbb{R}$, $f(x)=7-x$. Then $f(f(x))=7-(7-x)=x$, so f is its own inverse.

10.4 Inverse of a composition

$$(g\circ f)^{-1} = f^{-1}\circ g^{-1}$$

The order reverses: undo g first, then undo f .

EXAMPLE 10.4

Let $f(x)=x+2$ and $g(x)=3x$. Then $(g\circ f)(x)=3x+6$, whose inverse is $(x-6)/3$. Also $f^{-1}(g^{-1}(x))=x/3-2=(x-6)/3$.

11. Domain Restrictions and Inverse Graphs

The same formula can change type when its domain or codomain changes.

11.1 Domain and codomain determine properties

WORKED EXAMPLE 11.1

For $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, the function is not injective because $f(1) = f(-1)$, and not surjective because negative reals are not reached. Therefore it is not invertible.

11.2 Restricting the domain

Restricting x^2 to non-negative inputs removes the $\pm$ collision.

WORKED EXAMPLE 11.2

Let $f: [0, \infty) \rightarrow [0, \infty)$, $f(x) = x^2$. It is injective on non-negative inputs and every $y \geq 0$ has preimage $\sqrt{y}$. Hence f is bijective and $f^{-1}(x) = \sqrt{x}$.

EXTRA EXAMPLE - Negative branch

If instead $f: (-\infty, 0] \rightarrow [0, \infty)$, $f(x) = x^2$, then the inverse is $f^{-1}(x) = -\sqrt{x}$ because the preimage must remain non-positive.

11.3 Graph relationship

The graph of f^{-1} is the reflection of the graph of f in the line $y = x$. A point (a, b) on f becomes (b, a) on f^{-1} .

EXAMPLE 11.3

Since $(2, 7)$ lies on $f(x) = 3x + 1$, the reversed point $(7, 2)$ lies on $f^{-1}(x) = (x - 1)/3$.

EXAM TRAP

Swapping x and y may produce an **inverse relation** even when the original function is not bijective. For $y = x^2$ on $\mathbb{R}$, swapping gives $x = y^2$ and $y = \pm\sqrt{x}$, not a single-valued inverse function on $\mathbb{R}$.

12. Identity Functions and the Algebra of Inverses

The identity function leaves every element unchanged and acts like 1 does in multiplication.

12.1 Definition

$$i_A: A \rightarrow A, \quad i_A(x) = x$$

EXAMPLE 12.1

If $A = \{p, q, r\}$, then $i_A = \{(p, p), (q, q), (r, r)\}$. It is injective, surjective, bijective and self-inverse.

12.2 Identity under composition

$$\text{For } f: A \rightarrow B: i_B \circ f = f \text{ and } f \circ i_A = f$$

EXAMPLE 12.2

If $f(x) = 2x + 5$, then $(i \circ f)(x) = i(2x + 5) = 2x + 5$ and $(f \circ i)(x) = f(x) = 2x + 5$.

12.3 Inverses undo functions

$$f^{-1} \circ f = i_A \text{ and } f \circ f^{-1} = i_B$$

EXAMPLE 12.3

If $f: A \rightarrow B$ maps $1 \mapsto b$, $2 \mapsto c$, $3 \mapsto a$, then f^{-1} maps $b \mapsto 1$, $c \mapsto 2$, $a \mapsto 3$. Following f and then f^{-1} returns every element of A to itself.

12.4 Uniqueness of the inverse

A bijective function has exactly one inverse. If g and h both undo f , then $g = g \circ i_B = g \circ (f \circ h) = (g \circ f) \circ h = i_A \circ h = h$.

EXAMPLE 12.4

For $f(x) = x + 4$, any inverse must send y back to $y - 4$. No second, different function can undo f for every input.

13. Integrated Exam Examples

These examples combine the definitions and methods the way an assessment question usually does.

13.1 Integer linear function

QUESTION

Classify $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = 6x - 1$. State whether it is invertible.

Injective: $6u - 1 = 6v - 1$ implies $u = v$.

Surjective: $y = 6x - 1$ gives $x = (y + 1)/6$. For $y = 0$, $x = 1/6 \notin \mathbb{Z}$, so 0 is not reached.

Conclusion: injective only; not bijective; not invertible as $\mathbb{Z} \rightarrow \mathbb{Z}$.

13.2 Real linear function with inverse

QUESTION

For $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 7 - 2x$, prove bijectivity and find f^{-1} .

Injective: $7 - 2u = 7 - 2v$ gives $u = v$.

Surjective: for $y \in \mathbb{R}$ choose $x = (7 - y)/2 \in \mathbb{R}$, giving $f(x) = y$.

Inverse: $y = 7 - 2x$; swap and solve to get $f^{-1}(x) = (7 - x)/2$.

13.3 Composition plus classification

QUESTION

Let $f(x) = 2x - 1$ and $g(x) = x^2$, both $\mathbb{R} \rightarrow \mathbb{R}$. Find both compositions and classify f and g .

$(g \circ f)(x) = (2x - 1)^2$ and $(f \circ g)(x) = 2x^2 - 1$.

f is injective and surjective, hence bijective. g is neither: $g(1) = g(-1)$, and negative reals are missing.

13.4 Finite mapping

QUESTION

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $f = \{(1, a), (2, c), (3, b), (4, c)\}$. Find the range and classify f .

$\text{ran}(f) = \{a, b, c\} = B$, so f is surjective. It is not injective because $f(2) = f(4) = c$ with $2 \neq 4$. Hence it is surjective only and has no inverse function $B \rightarrow A$.

13.5 Restricted quadratic

QUESTION

Let $h: [2, \infty) \rightarrow [0, \infty)$, $h(x) = (x - 2)^2$. Is h invertible? Find its inverse.

On $x \geq 2$, $x - 2 \geq 0$, so h is strictly increasing and injective. Every $y \geq 0$ is reached by $x = 2 + \sqrt{y}$. Therefore h is bijective and $h^{-1}(x) = 2 + \sqrt{x}$.

14. Common Traps and How to Avoid Them

Mistake	Correction
Range = codomain	False in general. They are equal only for a surjective function.
A repeated output means it is not a function	False. It is still a function, but it is not injective.
No collisions proves surjective	No. No collisions proves injective.
Every codomain element reached proves injective	No. It proves surjective.
$g \circ f$ means $f(g(x))$	Wrong order. $g \circ f$ means $g(f(x))$; apply f first.
Composition is commutative	Usually false: $g \circ f$ and $f \circ g$ commonly differ.
Every function has an inverse function	False. Exactly the bijections are invertible.
f^{-1} means $1/f$	False. f^{-1} is the inverse function; $1/f$ is a reciprocal.
Swapping x and y always gives an inverse function	It may only give an inverse relation unless bijectivity holds.
The formula decides everything	False. Domain and codomain are part of the function.
If f and g are injective, $f \circ g$ and $g \circ f$ always both exist	False. Their domains and codomains must be compatible.
Checking a few outputs proves onto	Not for infinite sets. Use an arbitrary y proof or determine the full range.

15. Practice Test

Try these without looking at the solutions. The questions move from direct definitions to full exam-style proofs.

1. Let $f: \{1,2,3,4\} \rightarrow \{a,b,c,d\}$ be $1 \mapsto a, 2 \mapsto c, 3 \mapsto c, 4 \mapsto b$. Find $\text{ran}(f)$. Is f injective? Is it surjective?
2. Decide whether $R = \{(1,a), (1,b), (2,c)\}$ is a function from $\{1,2\}$ to $\{a,b,c\}$. Explain.
3. Find the range of $f: \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 5x + 2$.
4. Prove that $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -4x + 9$ is injective.
5. Prove that the same function in Question 4 is surjective.
6. Show that $q: \mathbb{R} \rightarrow \mathbb{R}, q(x) = x^2 - 1$ is not injective and not surjective.
7. Let $f(x) = x - 3$ and $g(x) = 2x + 5$. Find $g \circ f$ and $f \circ g$.
8. Let $f = \{(1,a), (2,b), (3,c)\}$ and $g = \{(a,4), (b,6), (c,5)\}$. Find $g \circ f$.
9. Find the domain of $(g \circ f)(x)$ when $f(x) = x + 1$ and $g(x) = 1/x$.
10. If $|A| = 8$ and $|B| = 6$, can a function $A \rightarrow B$ be injective? Can it be surjective?
11. If $|A| = |B| = 5$, how many bijections $A \rightarrow B$ exist?
12. Classify $f: \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 4x - 3$ as neither, injective only, surjective only or bijective.
13. Prove that $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 7$ is bijective and find f^{-1} .
14. Verify the inverse from Question 13 by both compositions.
15. Explain why $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ has no inverse function, then choose a restriction that makes it invertible.
16. Find the inverse of $f: [0, \infty) \rightarrow [3, \infty), f(x) = x^2 + 3$.
17. Determine whether $f(x) = 5 - x$ is self-inverse.
18. If f and g are injective, prove $g \circ f$ is injective.
19. If $g \circ f$ is surjective, what must be true about g ? Explain.
20. Let $f(x) = x + 2$ and $g(x) = 4x - 1$. Find $(g \circ f)^{-1}$ in two ways.

16. Fully Worked Solutions

Solution 1

$\text{ran}(f) = \{a, b, c\}$. It is not injective because 2 and 3 both map to c . It is not surjective because d is not reached. Therefore it is neither.

Solution 2

No. Input 1 has two outputs, a and b , so the relation violates the one-output-per-input rule.

Solution 3

$\text{ran}(f) = \{5k+2 \mid k \in \mathbb{Z}\}$, the integers congruent to 2 modulo 5.

Solution 4

Assume $f(u) = f(v)$. Then $-4u+9 = -4v+9$, so $-4u = -4v$ and $u = v$. Hence f is injective.

Solution 5

Let $y \in \mathbb{R}$. Solve $y = -4x+9$ to get $x = (9-y)/4 \in \mathbb{R}$. Then $f(x) = y$, so f is surjective.

Solution 6

Not injective: $q(1) = 0 = q(-1)$. Not surjective: $q(x) \geq -1$, so any $y < -1$, such as -2 , is missing.

Solution 7

$(g \circ f)(x) = g(x-3) = 2(x-3)+5 = 2x-1$. $(f \circ g)(x) = f(2x+5) = 2x+2$.

Solution 8

Follow the chains: $1 \rightarrow a \rightarrow 4$, $2 \rightarrow b \rightarrow 6$, $3 \rightarrow c \rightarrow 5$. Thus $g \circ f = \{(1,4), (2,6), (3,5)\}$.

Solution 9

$(g \circ f)(x) = 1/(x+1)$. Require $x+1 \neq 0$, so the domain is $\mathbb{R} \setminus \{-1\}$.

Solution 10

Injective is impossible because $8 > 6$. Surjective is possible because eight inputs are enough to reach six outputs.

Solution 11

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ bijections.

Solution 12

Injective: $4u - 3 = 4v - 3$ gives $u = v$. Not surjective: for $y = 0$, $x = 3/4 \notin \mathbb{Z}$. Therefore injective only.

Solution 13

Injective because $2u + 7 = 2v + 7$ implies $u = v$. Onto because for $y \in \mathbb{R}$, $x = (y - 7)/2$ is real and $f(x) = y$. Thus bijective. Inverse: $f^{-1}(x) = (x - 7)/2$.

Solution 14

$f^{-1}(f(x)) = ((2x + 7) - 7)/2 = x$, and $f(f^{-1}(x)) = 2((x - 7)/2) + 7 = x$.

Solution 15

It is not injective since $f(1) = f(-1)$. Restrict to $[0, \infty) \rightarrow [0, \infty)$, where it is bijective and the inverse is $\sqrt{x}$; or use $(-\infty, 0] \rightarrow [0, \infty)$, where the inverse is $-\sqrt{x}$.

Solution 16

Write $y = x^2 + 3$ with $x \geq 0$. Swap and solve: $x = y^2 + 3$ gives $y = \sqrt{x - 3}$. Thus $f^{-1}(x) = \sqrt{x - 3}$, domain $[3, \infty)$.

Solution 17

Yes. $f(f(x)) = 5 - (5 - x) = x$, so $f^{-1} = f$.

Solution 18

Assume $(g \circ f)(u) = (g \circ f)(v)$. Then $g(f(u)) = g(f(v))$. Injectivity of g gives $f(u) = f(v)$, and injectivity of f gives $u = v$. Thus $g \circ f$ is injective.

Solution 19

g must be surjective onto the final codomain. Every final value reached by $g \circ f$ is, in particular, an output of g .

Solution 20

Directly: $(g \circ f)(x) = 4(x + 2) - 1 = 4x + 7$, so its inverse is $(x - 7)/4$. By the rule: $f^{-1}(x) = x - 2$ and $g^{-1}(x) = (x + 1)/4$; then $f^{-1}(g^{-1}(x)) = (x + 1)/4 - 2 = (x - 7)/4$.

17. Final Memory Sheet and Mastery Checklist

17.1 One-page concept map

Concept	Exact statement	Memory cue
Function	Every domain element has exactly one output	One output per input
Range	$\text{ran}(f) = \{f(x) x \in A\}$	Actual outputs
Injective	$f(u) = f(v) \Rightarrow u = v$	No collisions
Surjective	$\text{ran}(f) = B$	No empty spots
Bijjective	Injective and surjective	Perfect matching
Composition	$(g \circ f)(x) = g(f(x))$	Right-hand function first
Invertible	$f \text{ invertible} \Leftrightarrow f \text{ bijective}$	Arrows reverse safely
Identity	$i_A(x) = x$	Changes nothing

17.2 Proof templates

INJECTIVITY

Assume $f(u) = f(v)$. Simplify until $u = v$. If you want to disprove injectivity, find $u \neq v$ with $f(u) = f(v)$.

SURJECTIVITY

Take arbitrary y in the codomain. Solve $y = f(x)$, then prove the resulting x lies in the domain. To disprove onto, exhibit one missing codomain value.

INVERSE

First prove bijectivity. Then swap x and y , solve for y , reverse domain/codomain, and verify both compositions if asked.

17.3 How to earn full proof marks

- Name arbitrary elements clearly: let $u, v \in A$ or let $y \in B$.
- Write the definition you are using before the algebra.
- For surjectivity, explicitly confirm that the chosen preimage belongs to the domain.
- A counterexample must satisfy every stated domain/codomain condition.
- End with a conclusion that answers the exact question: injective, onto, bijective or invertible.

17.4 Mastery checklist

- ☐ I can distinguish a relation from a function.
- ☐ I can identify domain, codomain, image, preimage and range.
- ☐ I can determine the range of finite and algebraic functions.
- ☐ I can define and recognise injective, surjective and bijective functions.
- ☐ I can prove injectivity and surjectivity formally.
- ☐ I can disprove either property with a valid counterexample.
- ☐ I can calculate compositions from formulas and ordered pairs.
- ☐ I check composition compatibility and domain restrictions.
- ☐ I know composition is associative but generally not commutative.
- ☐ I know which properties are preserved under composition.
- ☐ I can use finite cardinality shortcuts.
- ☐ I understand inverse relations and why bijectivity is necessary.
- ☐ I can find and verify inverse functions.
- ☐ I can restrict a domain to make a function invertible.
- ☐ I understand identity functions and the inverse laws.
- ☐ I can solve integrated exam questions without mixing up injective and surjective.

17.5 Rapid self-test

ANSWER THESE ALOUD

1. What is the difference between range and codomain?
2. Which property means no collisions?
3. Which property means no missing codomain values?
4. In $g \circ f$, which function is applied first?
5. When does an inverse function exist?
6. What two compositions verify an inverse?
7. Why can the same formula have different properties on different domains?

SEVEN-LINE ULTRA-QUICK REVISION

RANGE = what actually comes out
SURJECTIVE = no empty codomain spots
INJECTIVE = no collisions
BIJECTIVE = no empty spots + no collisions
COMPOSITION = right-hand function first
INVERTIBLE = bijective
IDENTITY = the input returns unchanged

END OF UNIT 7